



復旦大學

Backward Analysis of the Total Least Squares and its Randomized Algorithms

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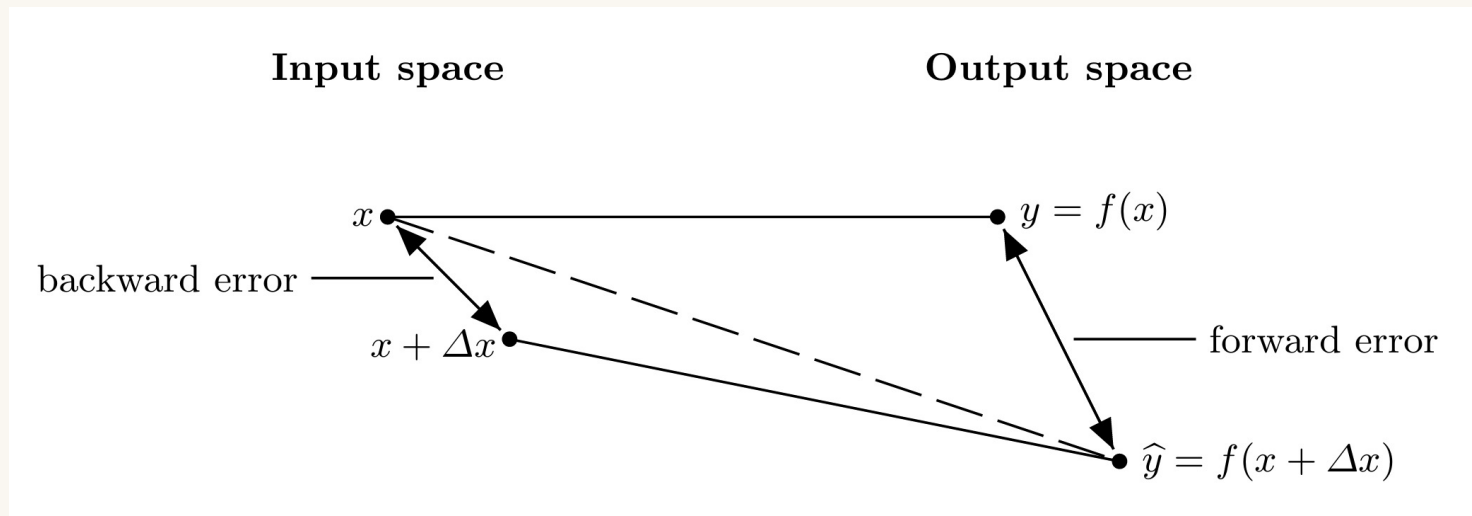
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Question: Why do we need the backward error analysis in general?

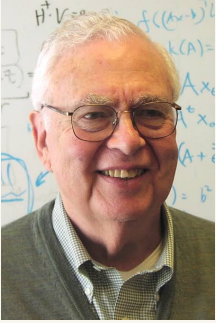
Backward stability:

A method for computing $y = f(x)$ is called *backward stable* if, for any x , it produces a computed $\hat{y}$ with a small backward error, that is, $\hat{y} = f(x + \Delta x)$ for some small Δx ('backward error'). The definition of 'small' will be context-dependent. (Higham, 2002)



Depending on **the difficulty of the problem itself**, it could be hard to achieve ideally small forward error, that is, we can only obtain **limited accuracy** in the real-world! → **Backward stability is the most we can expect for a solver.**

Question: Why are we interested in the backward error analysis of the TLS?



G. H. Golub and C. F. Van Loan, *An analysis of the total least squares problem*, SIAM Journal on Numerical Analysis, 17 (1980), pp. 883–893.

Van Huffel, Sabine, ed. *Recent advances in total least squares techniques and errors-in-variables modeling*, (1997).



Category	Least squares (LS)	Total least squares (TLS)
Formulation	$\min_x \ Ax - b\ _2$	$\min_{\Delta A, \Delta b, x} \ [\Delta A, \Delta b]\ _F, \text{ s.t. } (A + \Delta A)x = b + \Delta b$
Normal equation	$A^T(b - Ax) = 0$	$(A^T A - \sigma'_{n+1}{}^2 I)x = A^T b$
Direct method	QR: $x = R^\dagger Qb$	SVD: $x = -\frac{v'_{n+1}[1:n]^T}{v'_{n+1}[n+1]}$, $B \equiv [A, b] = U\Sigma V^T$.
More techniques	Sketching-and-solve; Iterative refinement; Randomized preconditioning;...	 <i>(relatively underdeveloped)</i>

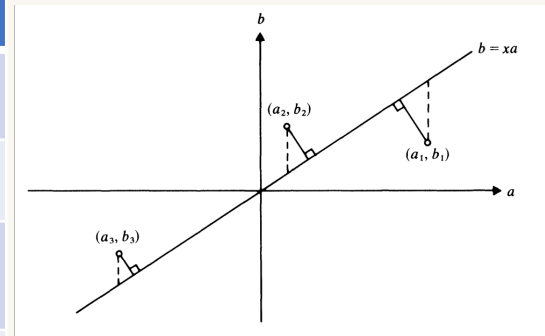
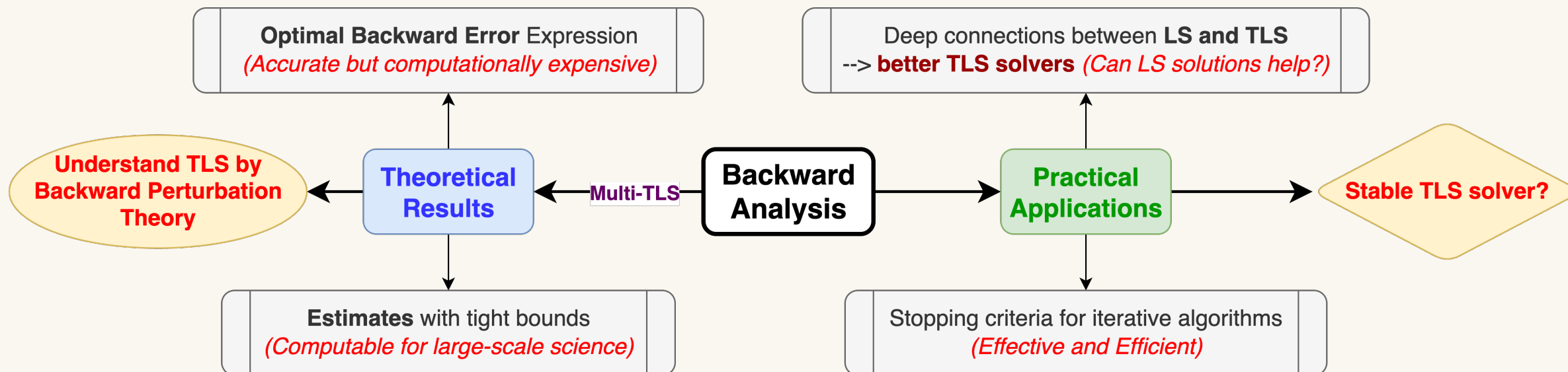
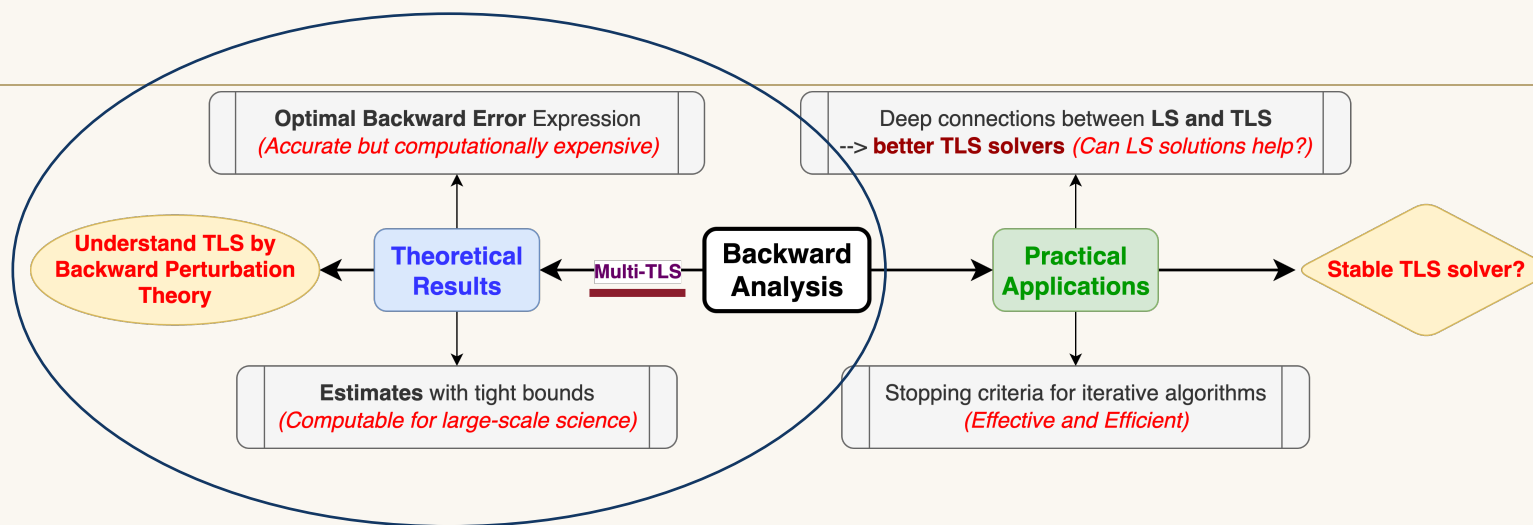


Fig: Least squares (--) VS
Total least squares (—)

Notation: Denote the singular values of B as $\sigma'_i (i = 1, \dots, n+1)$ and those of A as $\sigma_k (k = 1, \dots, n)$.
Denote the right singular vectors of B as $v'_i (i = 1, \dots, n+1)$ and those of A as $v_k (k = 1, \dots, n)$.

Outline





01 Optimal backward error and its efficient estimate

Main contributions of this section:

- I. theoretically complete the TLS backward analysis
- II. efficiently compute backward error for any given TLS solution

Backward Perturbation for the multidimensional TLS (mTLS)

(mTLS) Problem formulation: $\min_{\Delta A, \Delta B} \|\Delta A, \Delta B\|_F$, subject to $(A + \Delta A)X = B + \Delta B$, where $A \in \mathbb{R}^{m \times n}$, $m \gg n$.

Given a computed solution $\tilde{X} \in \mathbb{R}^{m \times n}$, the **θ -weighted backward error** of the mTLS problem is defined as

$$\eta_{\theta}(\tilde{X}) = \min \{ \|\Delta A, \Delta B\|_F : \tilde{X} \text{ is the true solution to a nearby mTLS system } (A + \Delta A)\tilde{X} = B + \Delta B \}, \text{ i.e.,}$$

$$\min_{\Delta A, \Delta B} \|\Delta A, \Delta B\|_F, \text{ subject to } (A + \Delta A)\tilde{X} = B + \Delta B.$$

GOALS: ① *Theoretically* establish the backward perturbation theory for the general mTLS problem;

→ give the explicit expression of $\eta_{\theta}(\tilde{X})$.

② *Practically* develop efficient algorithms for the computation of $\eta_{\theta}(\tilde{X})$.

→ derive estimates with tight theoretical bounds and exploit randomization in the algorithm design.

Backward Perturbation Theory – Related works

Least Squares (LS) Problem —

• **Single right-hand side** : $\min_x \|Ax - b\|_2$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$.

-- Given a nonzero $\hat{x} \in \mathbb{R}^n$ with $\hat{r} = b - A\hat{x}$, the following two sets are identical (*Walden, Karlson and Sun, 1995*),

$$\mathcal{E} = \{E \in \mathbb{R}^{m \times n} : \|b - (A + E)\hat{x}\| = \min_{x \in \mathbb{R}^n} \|b - (A + E)x\|\}, \mathcal{M} = \{\hat{r}\hat{x}^\dagger - vv^\dagger(A + \hat{r}\hat{x}^\dagger) + (I - vv^\dagger)Z(I - \hat{x}\hat{x}^\dagger) : v \in \mathbb{R}^m, Z \in \mathbb{R}^{m \times n}\}.$$

-- Define $\lambda = \lambda_{\min}\left(AA^\top - \frac{\hat{r}\hat{r}^\top}{1 + \|\hat{x}\|^2}\right)$ and $\mathcal{G} = \{[E, f] : \|(b + f) - (A + E)\hat{x}\| = \min_{x \in \mathbb{R}^n} \|(b + f) - (A + E)x\|\}$, then

$$\mu(\hat{x}) = \min_{[E, f] \in \mathcal{G}} \|[E, f]\|_F = \begin{cases} \frac{\|\hat{r}\|}{\|\hat{x}\|}, & \text{if } \lambda \geq 0, \\ \left(\frac{\|\hat{r}\|^2}{\|\hat{x}\|^2} + \lambda\right)^{\frac{1}{2}}, & \text{if } \lambda < 0. \end{cases} \quad (\text{Walden and Karlson, 1997}) \rightarrow \text{Computationally expensive and unstable}$$

-- Stable alternative: $\mu(\hat{x}) = \min\left\{\frac{\|\hat{r}\|}{\|\hat{x}\|}, \sigma_{\min}[A, B]\right\}$, $B = \frac{\|\hat{r}\|}{\|\hat{x}\|}(I - \hat{r}\hat{r}^\dagger)$ (*Higham, 2002*)

Efficient estimate: $\tilde{\mu}(\hat{x}) = \left\|(\|\hat{x}\|^2 AA^\top + \|\hat{r}\|^2 I)^{-\frac{1}{2}} A^\top \hat{r}\right\|$ (*Walden and Karlson, 1997; Gu, 1998; Gratton, 2012; Grcar et al., 2008*)

• **Multiple right-hand side** (*Hallman, 2020*)



Backward Perturbation Theory – Related works

Total Least Squares (TLS) Problem —

· **Single right-hand side** : (Scaled TLS) $\min_{\Delta A, \Delta b} \|[\Delta A, \gamma \Delta b]\|_F$, subject to $(A + \Delta A)x = b + \Delta b$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $m \gg n$.

-- The θ -weighted backward error for a computed solution $\tilde{x} \in \mathbb{R}^n$ is

$$\min_{E, f} \| [E, \theta f] \|_F, \text{ subject to } \tilde{x} \text{ solves a nearby scaled TLS problem } (A + E)x \approx b + f.$$

-- Denote $\tilde{r} = b - A\tilde{x}$ and $M = A(I - \tilde{x}\tilde{x}^\top)A^\top - \frac{\theta^2 \tilde{r}\tilde{r}^\top}{1 + \theta^2 \|\tilde{x}\|^2} + \frac{\theta^2 (A\tilde{x} + \gamma^2 \|\tilde{x}\|^2 b)(A\tilde{x} + \gamma^2 \|\tilde{x}\|^2 b)^\top}{\theta^2 \|\tilde{x}\|^2 + \gamma^4 \|\tilde{x}\|^4}$, then (*Chang et al., 2009*)

$$\min_{E, f} \| [E, \theta f] \|_F^2 = \begin{cases} \frac{\theta^2 \|\tilde{r}\|^2}{1 + \theta^2 \|\tilde{x}\|^2}, & \text{if } \lambda_{\min}(M) \geq 0, \\ \frac{\theta^2 \|\tilde{r}\|^2}{1 + \theta^2 \|\tilde{x}\|^2} + \lambda_{\min}(M), & \text{if } \lambda_{\min}(M) < 0. \end{cases} \quad \rightarrow \text{Though explicit, not easy to understand by intuition}$$

· **Multiple right-hand side** (*Shan and Wei, 2026*)

→ Single right-hand side is naturally the special case : Accurate, explicit and intuitively right! Deeply connected with LS.

mTLS problem from an optimization perspective

The problem $\min_{\Delta A, \Delta b} \|\Delta A, \Delta b\|_F$, subject to $(A + \Delta A)x = b + \Delta b$ is equivalent to the $\operatorname{argmin}_y f(y) = \frac{\|b - Ay\|^2}{1 + \|y\|^2}$. → mTLS?

Theorem 1: $X \in \mathbb{R}^{n \times d}$ is the true solution to the mTLS problem

$$\min_{\Delta A, \Delta B} \|\Delta A, \Delta B\|_F, \text{ subject to } (A + \Delta A)X = B + \Delta B,$$

if it minimizes $f(X) = \left\| (B - AX)(X^\top X + I)^{-\frac{1}{2}} \right\|_F^2$, where $(X^\top X + I)^{-\frac{1}{2}}$ is the square root of the positive-definite matrix $X^\top X + I$.

In particular, X satisfies $A^\top(B - AX) = -X(X^\top X + I)^{-1}(B - AX)^\top(B - AX)$, or $(A + BX^\top)^\top(B - AX) = 0$.

Important takeaway: We can characterize the perturbations to the mTLS systems in an elegant way —

$$\mathcal{F}_{\text{mTLS}+} = \{[E, F] \in \mathbb{R}^{m \times (n+d)} : ((A + E) + (B + F)X^\top)^\top((B + F) - (A + E)X^\top) = 0\}$$

→ Why the relaxed feasible set could be useful? We shall expect the computed solution as a reasonable approximation to the true one.

Computed solution $\bar{X}$ with full column-rank

Theorem 2 (characterization of the perturbation to A when the right-hand side is fixed): Given $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times d}$, $\bar{X} \in \mathbb{R}^{n \times d}$ with full column-rank, then the following two sets are identical,

$$\begin{aligned}\mathcal{E}_{\text{mTLS}} &= \{E \in \mathbb{R}^{m \times n}: ((A + E) + B\bar{X}^\top)^\top [B - (A + E)\bar{X}] = 0\}, \\ \mathcal{M}_{\text{mTLS}} &= \{-YY^\dagger(\mathbf{A} + \mathbf{B}\bar{X}^\top) + (I - YY^\dagger)[\hat{R}\bar{X}^\dagger + Z(I - \bar{X}\bar{X}^\dagger)]: Y \in \mathbb{R}^{m \times d}, Z \in \mathbb{R}^{m \times n}\}, \text{ with } \hat{R} = B - A\bar{X}.\end{aligned}$$

Comparison with mLS — Deep connections with Least Squares!

Given $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times d}$, $\hat{X}_{\text{LS}} \in \mathbb{R}^{n \times d}$ with full column-rank, then the following two sets are identical (Sun, 1996),

$$\begin{aligned}\mathcal{E}_{\text{LS}} &= \{E \in \mathbb{R}^{m \times n}: \|(A + E)\hat{X}_{\text{LS}} - B\|_F = \min_{X \in \mathbb{R}^{n \times d}} \|(A + E)X - B\|_F\}, \\ \mathcal{M}_{\text{LS}} &= \{-YY^\dagger \mathbf{A} + (I - YY^\dagger)[\hat{R}_{\text{LS}}\hat{X}_{\text{LS}}^\dagger + Z(I - \hat{X}_{\text{LS}}\hat{X}_{\text{LS}}^\dagger)]: Y \in \mathbb{R}^{m \times d}, Z \in \mathbb{R}^{m \times n}\}, \text{ with } \hat{R}_{\text{LS}} = B - A\hat{X}_{\text{LS}}.\end{aligned}$$

Theoretical explicit optimal backward error for mTLS

Recall that $\eta_\theta(\tilde{X}) = \min \{ \|[E, \theta F]\|_F : \tilde{X} \text{ is the true solution to a nearby mTLS system } (A + E)X \approx B + F \}$, i.e.,

$$\min_{\Delta A, \Delta B} \|\Delta A, \Delta B\|_F, \text{ subject to } (A + E + \Delta A)\tilde{X} = B + F + \Delta B.$$

Theorem 3 (optimal backward error with all data perturbed): Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times d}$, $\tilde{X} \in \mathbb{R}^{n \times d}$ with full column-rank. Define

$\tilde{A} = (A + B\tilde{X}^\top)(I + \theta^{-2}\tilde{X}\tilde{X}^\top)^{-\frac{1}{2}}$ and $\tilde{R} = \theta(B - A\tilde{X})(I + \theta^2\tilde{X}^\top\tilde{X})^{-\frac{1}{2}}$, where θ is the positive weighting parameter. Write $\lambda_1^{(\theta)} \leq \lambda_2^{(\theta)} \leq$

$\dots \leq \lambda_m^{(\theta)}$ as the eigenvalues of the matrix $\tilde{A}\tilde{A}^\top - \tilde{R}\tilde{R}^\top$. Then, the optimal backward error is $\eta_\theta(\tilde{X}) = \left[\|\tilde{R}\|_F^2 + \lambda_*^{(\theta)} \right]^{\frac{1}{2}}$, where

$$\lambda_*^{(\theta)} = \begin{cases} 0, & \text{if } 0 \leq \lambda_1^{(\theta)}, \\ \lambda_1^{(\theta)} + \lambda_2^{(\theta)} + \dots + \lambda_s^{(\theta)}, & \text{if } \lambda_s^{(\theta)} < 0 \leq \lambda_{s+1}^{(\theta)} \text{ with } s < d, \\ \lambda_1^{(\theta)} + \lambda_2^{(\theta)} + \dots + \lambda_d^{(\theta)}, & \text{if } \lambda_d^{(\theta)} < 0. \end{cases}$$

* For the rank-deficient case & limited data-perturbation case, please go to our original paper.

Practical computation of the backward error (Single right-hand side)

Corollary 4 (Optimal backward error for a TLS): By setting the parameter $d = 1$, we have the optimal backward error for the TLS system with a single right-hand side as

$$\tau_{\theta}(\tilde{x}) = \min_{E, f} \|[E, \theta f]\|_F = \left[\|\tilde{r}\| + \min \{0, \lambda_{\min}(\tilde{A}\tilde{A}^{\top} - \tilde{r}\tilde{r}^{\top})\} \right]^{\frac{1}{2}},$$

where $\tilde{A} = (A + b\tilde{x}^{\top})(I + \theta^{-2}\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}$ and $\tilde{r} = \theta(I + \theta^2\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}(b - A\tilde{x})$.

For the numerical stability, it should be computed by

$$\min \{\|\tilde{r}\|, \sigma_{\min}(M)\}, \text{ with } M = \begin{pmatrix} \tilde{A}^{\top} \\ \|\tilde{r}\|(I - \tilde{r}\tilde{r}^{\dagger}) \end{pmatrix}.$$

Theorem 5 (Computable estimate for a TLS): Define

$$\tilde{\tau}_{\theta}(\tilde{x}) = \left\| (\tilde{A}^{\top}\tilde{A} + \|\tilde{r}\|^2 I)^{-\frac{1}{2}} \tilde{A}^{\top} \tilde{r} \right\|,$$

where $\tilde{A} = (A + b\tilde{x}^{\top})(I + \theta^{-2}\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}$ and $\tilde{r} = \theta(I + \theta^2\tilde{x}\tilde{x}^{\top})^{-\frac{1}{2}}(b - A\tilde{x})$, then $\tilde{\tau}_{\theta}(\tilde{x}) \leq \tau_{\theta}(\tilde{x}) \leq \sqrt{2}\tilde{\tau}_{\theta}(\tilde{x})$.

Practical computation with Sketching techniques

Definition 6 (Sketching matrix): A random matrix $S \in \mathbb{R}^{s \times m}$ is called a sketching matrix for a matrix $A \in \mathbb{R}^{m \times n}$ with distortion $\epsilon \in (0,1)$, if

$$\forall y \in \mathbb{R}^n, (1 - \epsilon)\|Ay\| \leq \|S(Ay)\| \leq (1 + \epsilon)\|Ay\|.$$

For probabilistic sketching matrices, they satisfy the property with high probability.

Lemma 7 (Randomized preconditioning): Let S be a sketching matrix for a full-rank matrix A with distortion $\epsilon \in (0,1)$.

Consider a factorization $SA = QK$, where Q is orthogonal and K is square. Then,

$$\frac{1}{1+\epsilon} \leq \sigma_{\min}(AK^{-1}) \leq \sigma_{\max}(AK^{-1}) \leq \frac{1}{1-\epsilon}. \text{ (Woolfe et al. , 2008)}$$

Practical computation with Sketching techniques

Theorem 8 (Computable estimate for a TLS with sketching): Let $S \in \mathbb{R}^{m \times (n+1)}$ be a sketching matrix for $\text{span}([A, b])$ and $SA = QK$ as above. Write $\tilde{A} = (A + b\tilde{x}^\top)(I + \theta^{-2}\tilde{x}\tilde{x}^\top)^{-\frac{1}{2}}$ and $\tilde{r} = \theta(I + \theta^2\tilde{x}\tilde{x}^\top)^{-\frac{1}{2}}(b - A\tilde{x})$. Define $\tilde{\tau}_\theta^{sk}(\tilde{x}) = \left\| \left((S\tilde{A})^\top (S\tilde{A}) + \|\tilde{r}\|^2 I \right)^{-\frac{1}{2}} \tilde{A}^\top \tilde{r} \right\|$, then $(1 - \epsilon)\tilde{\tau}_\theta^{sk}(\tilde{x}) \leq \tau_\theta(\tilde{x}) \leq \sqrt{2}(1 + \epsilon)\tilde{\tau}_\theta^{sk}(\tilde{x})$.

Choice of sketching: sparse sign embedding $S^{d \times m} = \xi^{-\frac{1}{2}}[s_1, \dots, s_m]$, where s_i are independent random vectors with ξ nonzero entries (± 1 , both with probability $\frac{1}{2}$) placed in uniformly random positions. Choose $s \geq \text{const} \cdot \frac{n \log n}{\eta^2}$ and $\xi \geq \text{const} \cdot \frac{\log n}{\eta}$ ($s = 12n, \xi = 8$).

Algorithm 3.1 Efficient estimate for the optimal backward error of a TLS

Require: Matrix $A \in \mathbb{R}^{m \times n}$, right-hand side $b \in \mathbb{R}^m$, computed solution $\tilde{x} \in \mathbb{R}^n$, weighting parameter θ

Ensure: Sketched estimate $\tilde{\tau}_\theta^{sk}(\tilde{x})$ of the optimal backward error

- 1: Draw a sketching matrix $S \in \mathbb{R}^{s \times m}$ for $[A, b]$
- 2: $\bar{A} \leftarrow A + b\tilde{x}^\top$, $\tilde{r} \leftarrow \theta(1 + \theta^2\|\tilde{x}\|^2)^{-1/2}(b - A\tilde{x})$, $\zeta^2 \leftarrow \tilde{r}^\top \tilde{r}$
- 3: $\tilde{A} \leftarrow \bar{A} + \left(\bar{A} \cdot \left(\frac{\theta}{\sqrt{\theta^2 + \|\tilde{x}\|^2}} - 1 \right) \tilde{x} \right) \cdot \frac{\tilde{x}^\top}{\|\tilde{x}\|}$
- 4: $(\tilde{U}_{sk}, \tilde{\Sigma}_{sk}, \tilde{V}_{sk}) \leftarrow \text{svd}(S\tilde{A})$ ▷ $\tilde{\Sigma}_{sk}$ in column vector form
- 5: $\tilde{\tau}_\theta^{sk}(\tilde{x}) \leftarrow \left\| \left[\frac{1}{\sigma_i^2 + \zeta^2} \right]_{i=1}^n \odot (\tilde{V}_{sk}^\top \tilde{A}^\top \tilde{r}) \right\|$ ▷ Hadamard product $\odot$



Practical computation tests (Single right-hand side)

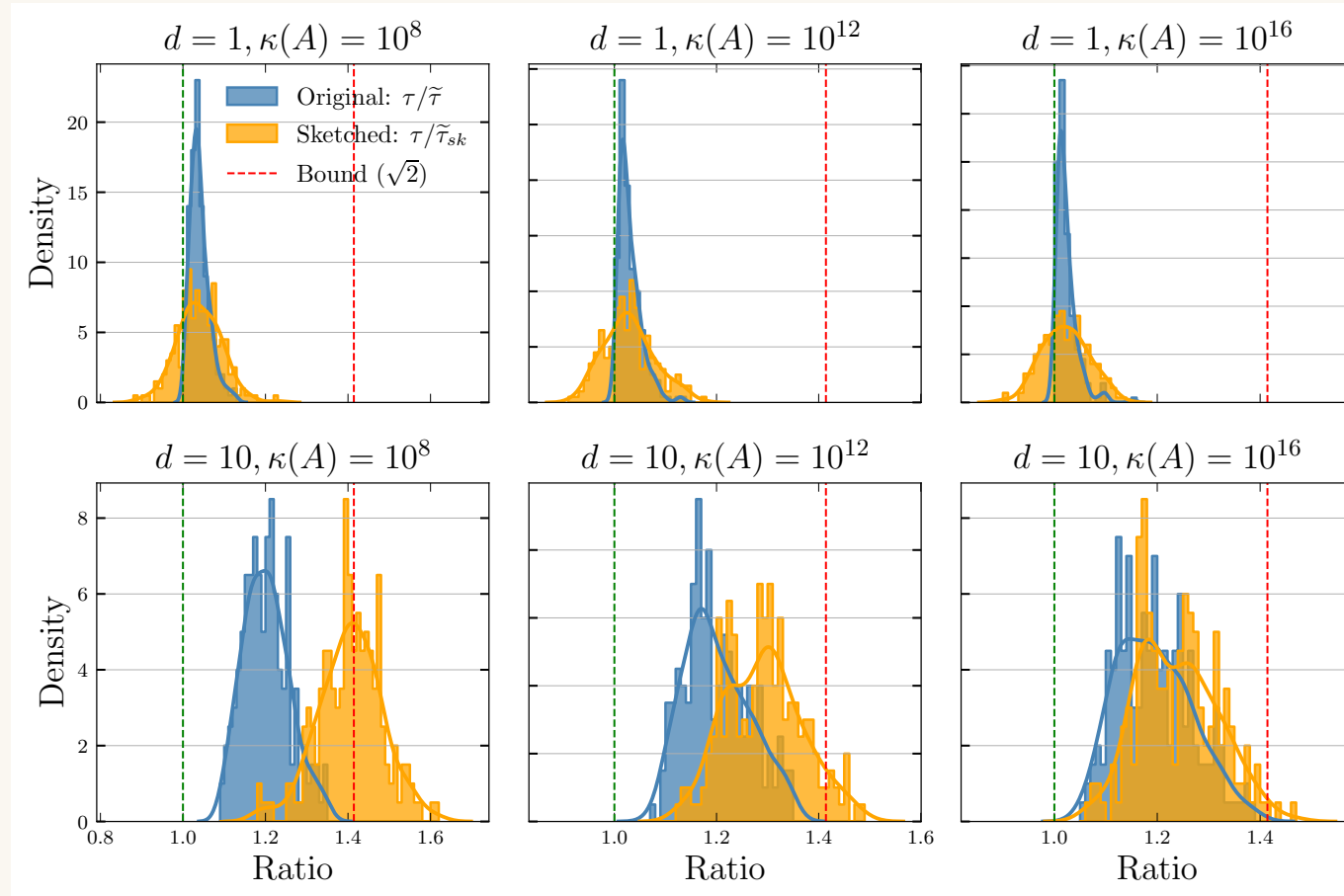


Figure: Tightness of the proposed estimations, with and without sketching techniques, for the optimal backward error of a total least squares. *Examples: fixed size $A^{3000 \times 20}$ with varying condition numbers.*

Practical computation of the backward error (Multiple right-hand sides)

Theorem 9 (Computable estimate for mTLS): Denote $\tilde{A} = (A + B\tilde{X}^\top)(I + \theta^{-2}\tilde{X}\tilde{X}^\top)^{-\frac{1}{2}}$ and $\tilde{R} = \theta(B - A\tilde{X})(I + \theta^2\tilde{X}^\top\tilde{X})^{-\frac{1}{2}}$. Let $r_{\tilde{A}}$ and $r_{\tilde{R}}$ be the ranks of $\tilde{A}$ and $\tilde{R}$, with SVD decompositions $\tilde{A} = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$ and $G = W\Lambda Z^\top$. Then,

$$(\text{recall that}) \eta_\theta(\tilde{X}) = [\|\tilde{R}\|_F^2 + \text{tr}(\tilde{A}\tilde{A}^\top - \tilde{R}\tilde{R}^\top)]^{\frac{1}{2}} \rightarrow \eta_\theta(\tilde{X}) = \eta(\tilde{A}, \tilde{R}) = \min \left\{ \|[M, N]\|_F : \tilde{A}^\top N + M^\top \tilde{R} = -\tilde{A}^\top \tilde{R} \text{ and } M^\top N = 0 \right\}.$$

Thus, we have a natural bound as $v^\theta(\tilde{X}) =$, which is equivalent to

$$v^\theta(\tilde{X}) = \left[\sum_{j=1}^{r_{\tilde{R}}} \lambda_j^2 \left\| (\tilde{A}\tilde{A}^\top + \lambda_j I)^{-\frac{1}{2}} \tilde{A}^\top w_j \right\| \right]^{\frac{1}{2}}.$$

It has a tight bound of $v_\theta(\tilde{X}) \leq \eta_\theta(\tilde{X}) \leq \sqrt{2}v_\theta(\tilde{X})$.

Similarly, using the sketching techniques, $v_\theta^{sk}(\tilde{X}) = \left[\sum_{j=1}^{r_{\tilde{R}}} \lambda_j^2 \left\| ((S\tilde{A})(S\tilde{A})^\top + \lambda_j I)^{-\frac{1}{2}} \tilde{A}^\top w_j \right\| \right]^{\frac{1}{2}}$ satisfies $1 - \epsilon \leq \frac{\eta_\theta(\tilde{X})}{v_\theta^{sk}(\tilde{X})} \leq \sqrt{2}(1 + \epsilon)$.

Practical computation tests (Multiple right-hand sides)

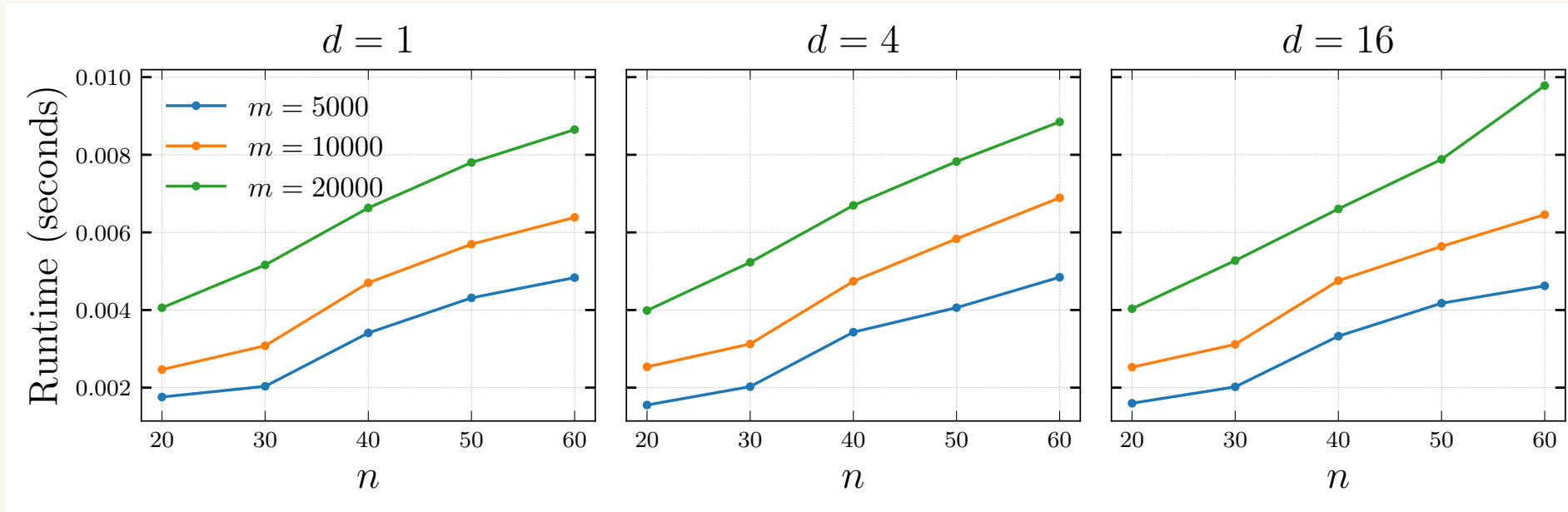
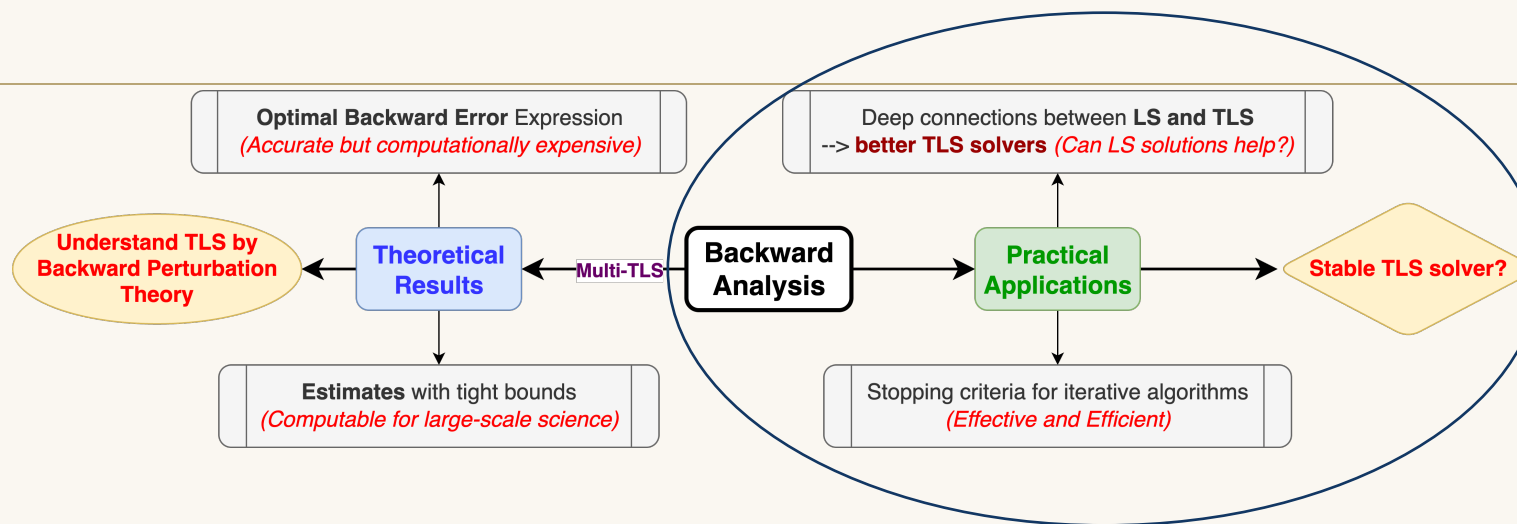


Figure: Efficiency of the sketched estimates for the optimal backward error of mTLS problems. *Examples: fixed-condition-number ($1e+6$) A with varying sizes, $m = [5000, 10000, 20000]$, $n = \text{linspace}(20, 60, 5)$, $d = [1, 4, 16]$.*



02 A fast randomized TLS solver – start from backward stability

Main contributions of this section:

- I. A showcase of the power of our theory
- II. An attempt to improve the TLS solver

Inspirations

What have we observed now?

The similarity between the optimal backward error of a same system through the lens of the least squares (LS) and the total least squares (TLS). The key difference lies in the additional ' $bx^\top$ ' part.

What could be the next try?

A backward stable LS solution may well be close to the true TLS solution. *This is intuitively accepted by lots of researchers but there has been little theoretical explanation before.*

What obstacles are expected?

The additional ' $bx^\top$ ' part plays a role and thus there must be some gaps.

Goal: to identify specific conditions under which the LS solution becomes advantageous for solving the TLS problem $Ax \approx b$ and can be computed more efficiently. → **Build the theory to bridge LS and TLS**



Proximity of the backward stable LS solutions

How ‘close’ is the backward stable LS solution $\hat{x}$ to the true TLS solution x_{TLS} ?

- Component-wise error with respect to the right singular vectors v_i of A ;
- Projection error: accurate in almost every component;
- The true LS residual $r_{\text{LS}} = b - Ax_{\text{LS}}$ matters: the challenges from the incompatible systems;
- Optimal perturbation to the nearest compatible system: already moderate.

Notation: Denote the computed LS solution as $\hat{x}$, the true LS solution as x_{LS} , the computed TLS solution as $\tilde{x}$ and the true TLS solution as x_{TLS} . Write u as the machine epsilon ($\approx 10^{-16}$ in double precision) and κ as the normal condition number of A . Always assume that $\kappa u \ll 1$, namely, A is numerically full-rank.



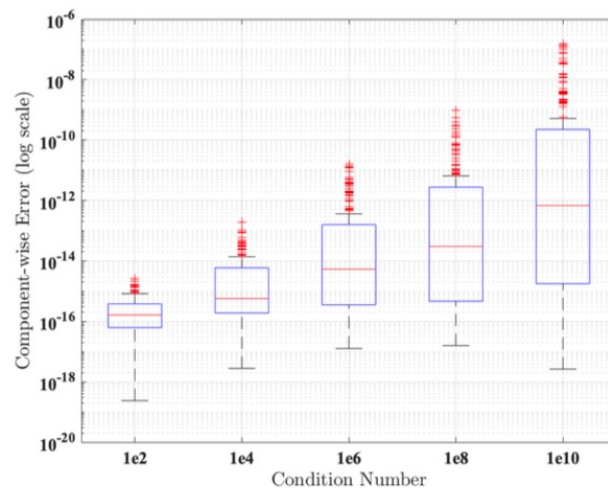
Backward stability makes the LS solution close to the true TLS solution

(LS) Backward stability of $\hat{x}$: $\hat{x} = \min_x \|(b + \Delta b) - (A + \Delta A)x\|$, where $\|\Delta A\| \lesssim \|A\|u$, $\|\Delta b\| \lesssim \|b\|u$.

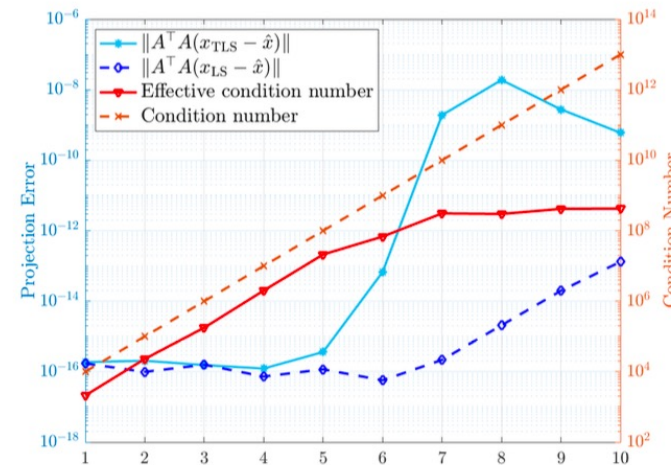
$$\Leftrightarrow v(\hat{x}) = \left\| (\|\hat{x}\|^2 A^\top A + \|b - A\hat{x}\|^2 I)^{-\frac{1}{2}} A^\top (b - A\hat{x}) \right\| \lesssim \|[A, b]\|u.$$

Theorem 10 (Component-wise error bound): For the right singular vectors v_i of A ($i = 1, 2, \dots, n$), we have

$$\left| v_i^\top \left(\left(1 - \frac{\sigma'_{n+1}{}^2}{\sigma_i^2} \right) x_{TLS} - \hat{x} \right) \right| \lesssim \frac{1}{\sigma_i} \|\hat{x}\| \|[A, b]\|u + \frac{1}{\sigma_i^2} \|b - A\hat{x}\| \|[A, b]\|u.$$



(a) Component-wise error



(b) Projection error

Backward stability makes the LS solution close to the true TLS solution

Proposition 11 (LS residual matters): Denote $\kappa = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$ as the normal condition number of A and $r_{\text{LS}} = b - Ax_{\text{LS}}$ be the true

LS residual. If there is a gap between 1 and $\kappa\|r\| < 1$, then $\frac{\sigma'_{n+1}}{\sigma_i}$ is away from 1.

Proposition 12 (Effective condition number): Denote $\kappa_{\text{eff}} = \frac{\|b\|}{\sigma_n\|x_{\text{LS}}\|}$, $\delta = \frac{\sigma'_{n+1}}{\sigma_i}$ and $\Delta x = x_{\text{TLS}} - x_{\text{LS}}$. Then we have

$$\frac{\|\Delta x\|}{\|x_{\text{LS}}\|} \leq \kappa_{\text{eff}} \frac{\delta}{1-\delta} \left(\sqrt{2} + \frac{\sigma_n}{\|b\|} \right).$$

Theorem 13 (Optimal perturbation norm): Given a backward stable LS solution $\hat{x}$, consider its nearest compatible system,

i.e., $\delta = \min \| [E, f] \|_{\text{F}}$, subject to $(A + E)\hat{x} = b + f$, then δ is upper-bounded as $\delta \lesssim \kappa u \frac{\|b\| + \|A\|\|\hat{x}\|}{(1 + \|\hat{x}\|^2)^{1/2}}$, where $\kappa u \ll 1$.

Randomized iterative methods – general framework

The normal equation for $Ax \approx b$ is $\begin{bmatrix} A^\top A & A^\top b \\ b^\top A & b^\top b \end{bmatrix} \begin{bmatrix} x \\ -1 \end{bmatrix} = \lambda \begin{bmatrix} x \\ -1 \end{bmatrix}$

to find $x = x_{\text{TLS}}$, $\lambda = \sigma'_{n+1}{}^2$.

Using the Rayleigh quotient iteration, then $\rho(x) = \frac{\|b - Ax\|^2}{1 + \|x\|^2}$.

Subroutine	Classic	Innovation
INITIALTOOL	Direct method	Fast backward LS solver (Epperly et al., 2024)
PCGSOLVER	QR / Cholesky	<i>Better preconditioners?</i>

Expensive cost

Algorithm 1.1 A general framework for the total least squares (TLS) problem

Require: Matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$, subroutine INITIALTOOL, subroutine PCGSOLVER

Ensure: TLS solution $x_{\text{TLS}} \in \mathbb{R}^n$

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1:  $x_0 \leftarrow \text{INITIALTOOL}(A, b)$                                 ▷ Construct the initial guess
2:  $r_0 = b - Ax_0$ 
3:  $\sigma_0^2 = r_0^\top r_0 / (x_0^\top x_0 + 1)$                                 ▷ Initial eigenvalue guess of  $B \equiv [A, b]$ 
4:  $u_0 \leftarrow \text{PCGSOLVER}(A^\top Au_0 = x_0)$                                 ▷ One step of the inverse iteration
5:  $x_1 = x_0 + \sigma_0^2 u_0$                                 ▷ Ready for RQI
6: for  $k = 1, 2, \dots$  do
7:    $r_k = b - Ax_k$ 
8:    $\sigma_k^2 = r_k^\top r_k / (x_k^\top x_k + 1)$ 
9:    $f_k = -A^\top r_k - \sigma_k^2 x_k$ 
10:   $g_k = -b^\top r_k + \sigma_k^2$ 
11:   $\omega_k \leftarrow \text{PCGSOLVER}((A^\top A - \sigma_k^2 I)\omega_k = -f_k)$ 
12:   $z_k = x_k + \omega_k$ 
13:   $\beta_k = (z_k^\top f_k - g_k) / (z_k^\top x_k + 1)$ 
14:   $u_k \leftarrow \text{PCGSOLVER}((A^\top A - \sigma_k^2 I)u_k = x_k)$ 
15:   $x_{k+1} = z_k + \beta_k u_k$ 
16: end for

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Sketching preconditioned conjugate gradient (SPCG)

Theorem 14 (Effectiveness of the sketching preconditioner): Take $\rho = \sigma'_{n+1}{}^2$ and assume

(1) $\eta \leq 0.9$ (useful sketching), $\|fl(SA) - SA\| \lesssim \|A\|u$ (stable sketching);

(2) $\kappa u \ll 1$ (A is numerically full-rank);

(3) $\frac{\sigma'_{n+1}}{\sigma_n}$ is away from 1,

then within the finite arithmetic, the condition numbers of

$$(1) R^{-\top} A^{\top} A R^{-1} x = y; (2) (R^{-\top} A^{\top} A R^{-1} - \rho R^{-\top} R^{-1}) x = y,$$

are both clustered around 1. Thus, the preconditioned systems are well-conditioned.

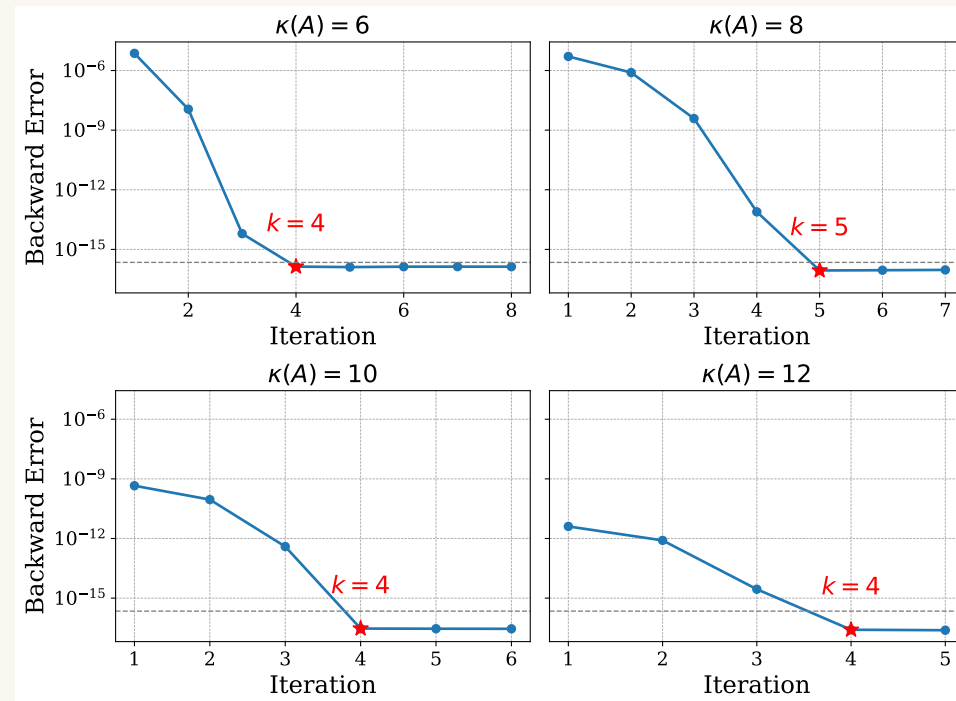
Proposition 15 (Convergence rate is maintained): Under the same assumptions, the inverse iteration is still of linear convergence. Or equivalently, the computing error does not influence the converging direction.

RQI-SPCG: Implementation details

Shift from inverse iteration to RQI loop: explicit formular (*Szyld, 1988*) but not applicable $\rightarrow$ empirical choices.

Termination criterion: Recall our backward analysis!

- ① **Classic:** an increase in $\gamma_k = ||f_k||/||\tilde{x}^{(k)}||$, where $\begin{pmatrix} f_k \\ g_k \end{pmatrix} = \begin{pmatrix} -A^T r^{(k)} - \rho_k x^{(k)} \\ -b^T r^{(k)} + \rho_k \end{pmatrix}$. For SPCG, a fixed number is enough.
- ② **Exploiting the backward error: advantageous**



RQI-SPCG: whole algorithm

Algorithm 1.3 RQI-SPCGTLS

Require: Matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$

Ensure: TLS solution $x_{\text{TLS}} \in \mathbb{R}^n$

1: $x_0 \leftarrow \text{FOSSIL}(A, b)$	$\triangleright \mathcal{O}(mn \log(n/u) + n^3)$
2: $r_0 = b - Ax_0$	$\triangleright 2mn + m$
3: $\sigma_0^2 = r_0^\top r_0 / (x_0^\top x_0 + 1)$	$\triangleright 2m + 2n - 2$
4: $u_0 \leftarrow \text{SPCG}(A, x_0, 0)$	$\triangleright \mathcal{O}(mn \log(n/u) + n^3)$
5: $x_1 = x_0 + \sigma_0^2 u_0$	$\triangleright 2n$
6: for $k = 1, 2, \dots$ do	
7: $r_k = b - Ax_k$	$\triangleright 2mn + m$
8: $\sigma_k^2 = r_k^\top r_k / (x_k^\top x_k + 1)$	$\triangleright 2m + 2n - 2$
9: $f_k = -A^\top r_k - \sigma_k^2 x_k$	$\triangleright 2mn + 2n$
10: $g_k = -b^\top r_k + \sigma_k^2$	$\triangleright 2m - 1$
11: $\omega_k \leftarrow \text{SPCG}(A, -f_k, \omega_k)$	$\triangleright \mathcal{O}(mn \log(n/u) + n^3)$
12: $z_k = x_k + \omega_k$	$\triangleright n$
13: $\beta_k = (z_k^\top f_k - g_k) / (z_k^\top x_k + 1)$	$\triangleright 4n - 2$
14: $u_k \leftarrow \text{SPCG}((A^\top A - \sigma_k^2 I)u_k = x_k)$	
15: $x_{k+1} = z_k + \beta_k u_k$	$\triangleright 2n$
16: end for	



Numerical Experiments – Synthetic data

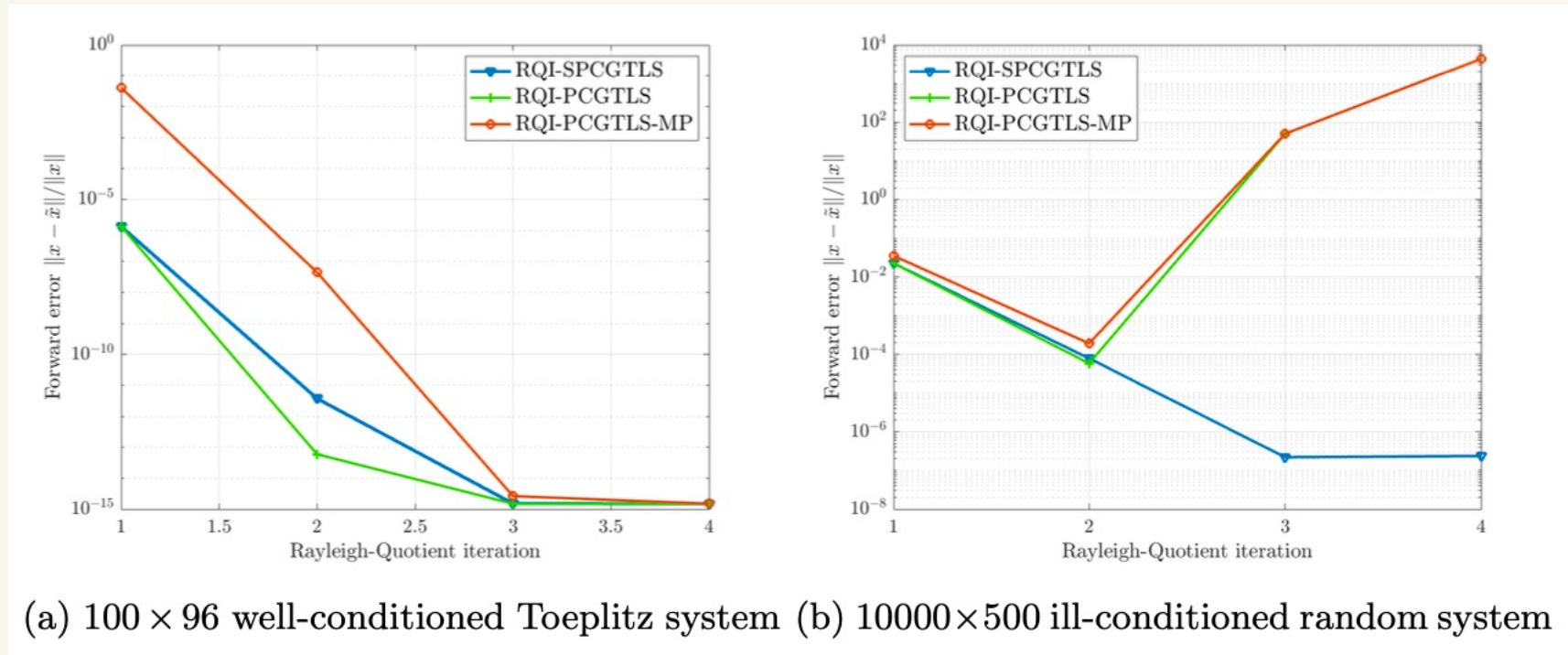


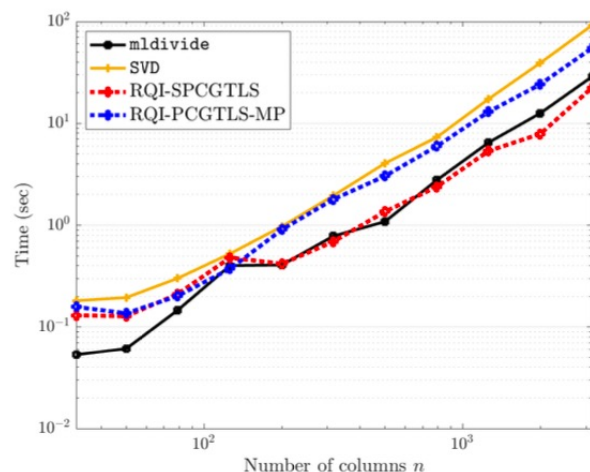
Figure: When the system tends to be overdetermined and ill-conditioned, the original or mixed-precision RQI-PCGTLS(-MP) fail to give good performance, while our RQI- SPCGTLS still works well.

Numerical Experiments – Real-world dataset

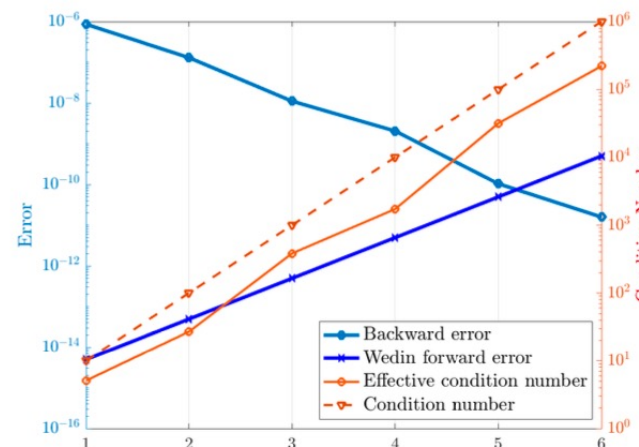
Matrix	m	n	nnz	SPCG	PCG-MP	Speedup
bibd_12_5	792	66	7920	0.02	0.03	1.5
bibd_17_8	24310	136	680680	0.04	0.09	2.3
bibd_20_10	184756	190	8314020	5.94	9.88	1.6
bibd_22_8	319770	231	8953560	6.38	20.72	3.3

SuiteSparse Matrix Collection;

T. A. Davis and Y. Hu, *The university of florida sparse matrix collection*, ACM Transactions on Mathematical Software (TOMS), 38 (2011), pp. 1–25.



(a) SUSY dataset



(b) Fairly compatible random systems

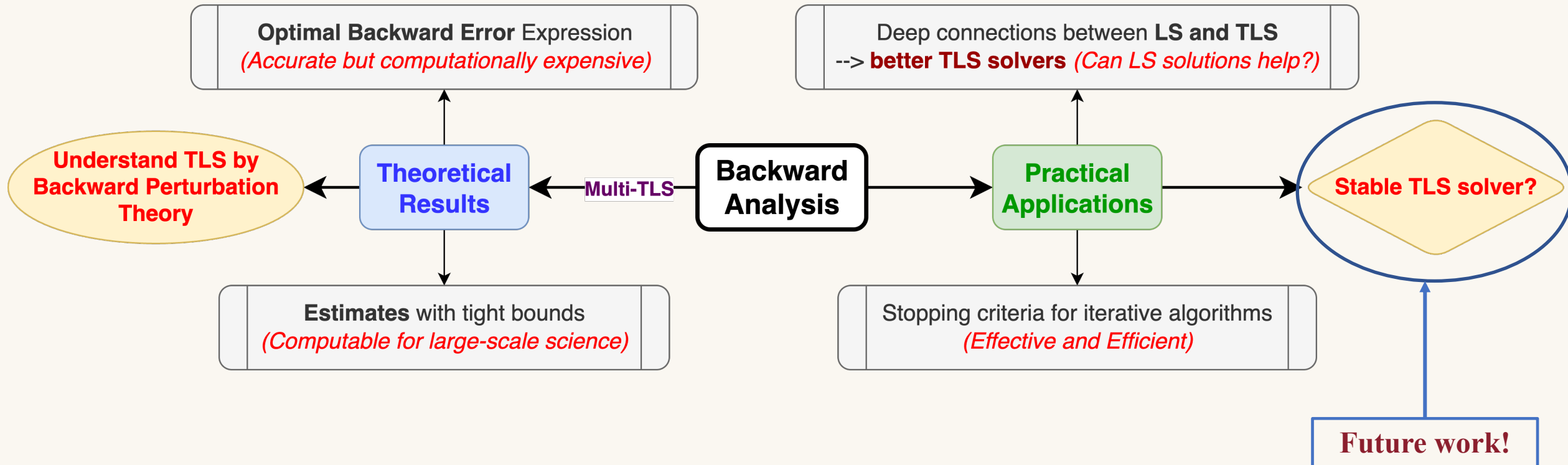
Figure: (a) is the runtime comparison for kernel ridge regression problems between the SVD, the RQI-PCGTLS-MP, and the RQI-SPCGTLS. (b) illustrates the backward error and the Wedin forward error by 500×20 systems with moderate residuals and changing condition numbers.

Kernel ridge regression(KKR):

$$K(z, z') = \exp\left(-\frac{\|z - z'\|^2}{2\sigma^2}\right),$$

with $\sigma = 4$.

Conclusion



Thank you!